>>>50 React Question

Basics of React:
1. What is React.js and what problem does it solve?
2. Describe the Virtual DOM and its role in React's rendering process.
3. How does React optimize the updating of components in the Virtual DOM?
4. Explain the difference between React components and HTML elements.
5. What is JSX? How does it differ from regular JavaScript?

Component Fundamentals:
6. What are functional components in React? How do they differ from class components?
7. Explain the component lifecycle methods in class components.
8. What is the purpose of the "constructor" method in a class component?
9. How do you pass data from a parent component to a child component?
10. What are props? How are they used to communicate between components?

State Management:
11. What is React state? How does it differ from props?
12. Explain how you would manage state in a functional component using the "useState" hook.
13. Describe the concept of "lifting state up" in React.
14. When would you use Redux for state management in React applications?
15. How does Redux work? Explain the basic principles of the Redux architecture.

Hooks:
16. List and briefly explain some of the built-in hooks in React.
17. What is the purpose of the "useEffect" hook? How can it be used to handle side effects?
18. How do you create custom hooks in React? Provide an example.
19. Explain the "useState" hook and provide an example of how you would use it.
20. What are the rules of using hooks in React?

Component Communication:
21. What is the Context API in React? How does it facilitate component communication?
22. When would you use the "useContext" hook in React applications?
23. Describe the process of prop drilling and how you can avoid it.
24. How can you achieve sibling component communication in React?
25. Explain the concept of event handling in React components.

Routing and Navigation:
26. What is React Router? How does it enable client-side routing?
27. How do you implement routing in a React application using React Router?
28. Explain the purpose of the "Link" and "NavLink" components in React Router.
29. What is dynamic routing? Provide an example of how you would implement it.
30. How would you handle "404 Not Found" scenarios in a React Router-based application?

Optimization and Performance:

31. What are some techniques for optimizing the performance of React applications?
32. Explain the concept of "memoization" and how it can improve performance.
33. When and why would you use the "shouldComponentUpdate" method in class components?
34. How can you use the "React DevTools" to diagnose and improve performance issues?
35. Describe the benefits of using the "key" prop when rendering lists of components.

Testing:
36. What are some popular testing libraries and frameworks used with React?
37. How would you write unit tests for React components using tools like Jest and React Testing Library?
38. Explain the difference between shallow rendering and full rendering in testing.
39. What is snapshot testing? How does it help in maintaining UI consistency?
40. How can you mock external dependencies and APIs while testing React components?

Styling:
41. Describe different approaches for styling React components.
42. What is CSS-in-JS? Provide an example of a CSS-in-JS library and how it's used.
43. How does the concept of "CSS modules" work in React applications?
44. Explain the purpose of the "styled-components" library and how it works.
45. Compare the advantages and disadvantages of using global styles vs. local styles in React.

Server Communication:
46. How would you make asynchronous API calls in a React application?
47. Describe the concept of "fetching data" and "component lifecycle" in relation to server communication.
48. When would you use the "axios" library for making API calls in React?
49. What is the purpose of the "componentDidMount" lifecycle method in relation to server communication?
50. How can you handle errors and loading states while fetching data in React?